

KYUNGMIN KIM

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RESEARCH INTERESTS

Database systems

EDUCATION

Seoul National University (SNU)

Mar. 2025 - Present

M.S. in Data Science

Sungkyunkwan University (SKKU)

Mar. 2016 - Feb. 2023

B.S. in Computer Science & Engineering

GPA: 4.05 / 4.5

EXPERIENCES

Database Systems Research Laboratory, Seoul National University

Mar. 2025 – Present

Graduate Research Assistant (Advisor: Prof. Sang-Won Lee)

Seoul, South Korea

- Research on PostgreSQL, Cloud-native database and Data lakes.

Kakao Corp.

Dec. 2021 - Feb. 2025

Data Engineer, Advertisement (AD) Development Team

Seongnam, South Korea

**Kakao offers KakaoTalk, the largest messenger application in South Korea with 45M+ users.*

- Build data pipelines to process large-scale real-time AD traffic using Apache Kafka and Flink, deployed on Hadoop and Kubernetes clusters.
- Ensure secure storage and fast querying of high-volume data across distributed storage systems like Elasticsearch and Cassandra.
- Develop data-providing APIs within a Microservice Architecture using Spring WebFlux, Java, and Apache Airflow.

Kakao Corp.

Summer, 2021

Software Engineer Intern, AD Development Team

Seongnam, South Korea

- Designed and implemented a prototype for a display AD system, including a demand-side platform, AD serving, billing, and reporting components.
- Focused on developing low-latency web servers with Java and Spring Boot and orchestrating data pipelines on Kubernetes with Kafka and ELK stacks for logging and analytics.

Robotics Research Laboratory, Purdue University

Aug. 2018 - Dec. 2018

Undergraduate Research Assistant (Advisor: Prof. Eric T. Matson)

West Lafayette, IN, United States

- Led project on developing an aerial defect-detection solution for solar panels using Python, drones, YOLO, and OpenCV.
- Created an algorithm to automatically identify abnormal solar cells by detecting contour shapes.

Database Systems Research Laboratory, Sungkyunkwan University

Mar. 2018 – Aug. 2019

Undergraduate Research Assistant (Advisor: Prof. Sang-Won Lee)

Suwon, South Korea

- Conducted performance analysis and optimization of database engines utilizing non-volatile memory as a storage device.
- Researched MySQL/InnoDB MVCC architecture and locking schemes to identify opportunities for enhancing database performance.

PUBLICATIONS

1. "Performance evaluation on concurrency control techniques of main-memory DBMS based on log device"
Kyungmin Kim, Jonghyeok Park, Sang-Won Lee
Korean DataBase Conference (KDBC 2022), Nov. 2022
2. "Analyzing the range of angles of a solar panel to detect defective cells, using a UAV"
Kyungmin Kim, Yoojin Choi, Keonyoung Shim, Hyewon Jeon, Jane Commerford, Eric T. Matson
Third IEEE International Conference on Robotic Computing (IRC 2019), Feb. 2019
3. "Analysis of MySQL/InnoDB Lock Contention on TPC-C Workload"
Kyungmin Kim, Jong-Hyeok Park, Mijin An, Sang-Won Lee
Korea Computer Congress (KCC 2019), Jul. 2019
4. "SQLite Database Engine with PMDK Library"
Jong-Hyeok Park, **Kyungmin Kim**, Gihwan Oh, Sang-Won Lee
Korean DataBase Conference (KDBC 2018), Oct. 2018
5. "Optimizing MySQL/InnoDB by Multi-Streamed data journal"
Soyee Choi, Dong Hyun Kang, **Kyung-Min Kim**, Sung-Youn Seo, Sang-Won Lee, Young Ik Eom
Korea Computer Congress (KCC 2018), Jul. 2018

AWARDS

SKKU Magna Cum Laude , Sungkyunkwan University	Feb. 2023
Dean's List , Sungkyunkwan University	Spring, 2016; Spring, 2017; Spring, 2019; Fall, 2021
Creativity Award in 2020 Army Entrepreneurship Contest , Republic of Korea Army	June. 2020
Excellence Paper Award , Korea Computer Congress (KCC)	Aug. 2019

SCHOLARSHIPS

K-SW Purdue Scholarship , South Korean Government (IITP)	Aug. 2018
Full Tuition Scholarship , Sungkyunkwan University	Spring, 2016 – Fall, 2022

PATENTS

Kyungmin Kim, Hyeonchul Yoon. "*Finding Music with Humming Sound.*" KR-Patent 10-2018-0046898

OTHER EXPERIENCES

Republic of Korea Army, Sergeant	Nov. 2019 - June. 2021
· Completed national duty with exemplary performance and leadership skills.	
Sungkyunkwan University Software Mentoring Program, Teacher	May. 2017 - Nov. 2017
· Taught Arduino programming to high school students.	
Edu Share Volunteering Program, Teacher	Sep. 2016 - May. 2017
· Taught mathematics to middle school students with few educational opportunities.	
Sungkyunkwan University Authorized Ambassador, Club Vice President	Jul. 2016 - Jul. 2017
· Promoted the university's Software department to prospective students.	

TECHNICAL SKILLS

Languages: Java, Kotlin, C/C++, Python, SQL
Frameworks: Apache Flink, Spring Boot, Spring Cloud
Environments: Apache Hadoop, Kubernetes, AWS
Database Technologies: MySQL, Elasticsearch, Apache Cassandra, Apache HBase, Redis
Developer Tools: Git, Docker, IntelliJ, Visual Studio Code